

Practice Lab:

DSE Search, Geospatial

- This Practice Lab is dependent on Discussion Module 7453, where most of the objects we create in this lab were introduced.
- This Practice Lab requires a working DSE system, with DSE Search enabled, and with ability to execute CQL DDL.
- If running the optional portions of this lab, this Practice Lab requires a ssh(C) prompt on all nodes operating DSE Search. A node reboot is required.
- All commands are run from CQLSH, or DSE Studio. CQLSH is required to COPY data into (a table).

Challenge 1: Closest Starbucks, USA



- 2017; 17,000 Starbucks locations USA
- Github downloaded data; 6000 US Starbucks locations, 10 years old ?
- Create the runtime to support the query; Where is my closest Starbucks ?

Challenge 1: Create basic runtime, Load Data-

- Given file; 7454_my_mapdata.small.csv

- CREAT KEYSPACE ...

- CREATE TABLE ...

- Load data;

```
COPY my_mapdata
```

```
(
```

```
md_pk,
```

```
...
```

```
md_type
```

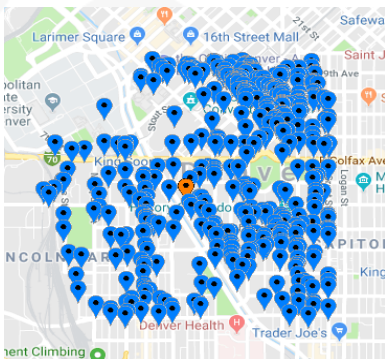
```
)
```

```
FROM '7454_my_mapdata.small.csv'
```

```
WITH HEADER = TRUE ;
```



Execute all of the instructions on
the notes page below-

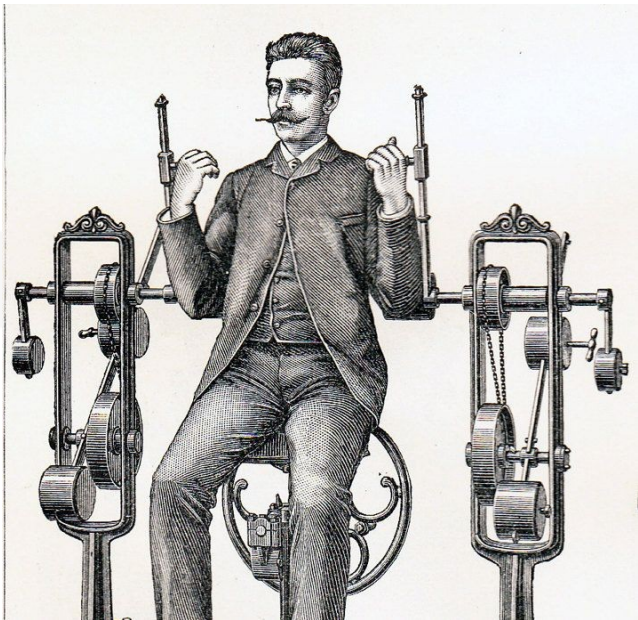


Go Farther: (Optional)



A navigator and explorer of African ancestry, Pedro Alonso Nino traveled with Christopher Columbus' first expedition to the New World in 1492. Pedro Nino was the pilot of Columbus' ship the "Santa Maria."
<http://legacysunfoldingjourney.blogspot.com/2013/07/african-explorers-and-settlers-of-new.html>

Challenge 2 (Optional): Add polygon support-



Extend (Copy) Challenge 1-

- Download JTS from, <http://central.maven.org/maven2/com/vivid/solutions/jts/1.13/DSE>
- Copy Jar into, ./resources/solr/lib (from parent directory of DSE).
- Restart DSE
- Add a polygon query; Intersects/POLYGON
- Use, SpatialRecursivePrefixTreeFieldType
- *To save time, you can do all of this work on new tables, etcetera.*

Practice Lab:

Lessons learned





End of Module: